



UNIVERSITY OF
WESTMINSTER 卐

BSc (Hons) Computer Science
SOFTWARE DEVELOPMENT I (4COSC006C)
ASSESSMENT (PART: A, B)
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Pseudocode

BEGIN

DECLARE tasks **AS LIST**

FUNCTION validate_date(date_str)

PARSE date_str **AS DATE FORMAT** "YYYY-MM-DD"

RETURN TRUE

END FUNCTION

FUNCTION add_task()

DISPLAY "Task Manager: Add Task"

REPEAT

DISPLAY "Enter task name: "

GET name

UNTIL name **IS NOT EMPTY**

DISPLAY "Enter task description: "

GET description

REPEAT

DISPLAY "Enter task priority (low, medium, high): "

GET priority

UNTIL priority = "low" OR priority = "medium" OR priority = "high"

REPEAT

DISPLAY "Enter due date (YYYY-MM-DD): "

GET due_date

UNTIL validate_date(due_date) = **TRUE**

DECLARE task **AS RECORD**

task.name

task.description

task.priority

task.due_date

END RECORD

SET task.name = name

SET task.description = description

SET task.priority = priority

SET task.due_date = due_date

ADD task **TO** tasks

DISPLAY "Task added successfully!"

END FUNCTION

```

FUNCTION view_tasks()
  IF tasks IS EMPTY THEN
    DISPLAY "No task(s) found."
    RETURN
  ENDIF

  DISPLAY "Task Manager: List of Tasks"

  SORT tasks BY PRIORITY HIGH TO LOW

  FOR task IN tasks DO
    DISPLAY "Name: " task.name
    DISPLAY "Description: " task.description
    DISPLAY "Priority: " task.priority
    DISPLAY "Due Date: " task.due_date
  END FOR
END FUNCTION

```

```

FUNCTION update_task()
  CALL view_tasks()
  DISPLAY "Task Manager: Update Task"

  REPEAT
    DISPLAY "Enter the task name to update: "
    GET task_name
  UNTIL task_name IS NOT EMPTY

  FOR task IN tasks DO
    IF task.name = task_name THEN
      DISPLAY "Current Description: " + task.description
      DISPLAY "Enter new description (Press Enter to Skip): "
      GET new_description
      IF new_description IS NOT EMPTY THEN
        SET task.description = new_description
      ENDIF

      DISPLAY "Current Priority: ", task.priority
      REPEAT
        DISPLAY "Enter new task priority (low, medium, high) (Press Enter to Skip): "
        GET new_priority
        IF new_priority IS EMPTY THEN
          BREAK
        ENDIF
      UNTIL new_priority = "low" OR new_priority = "medium" OR new_priority = "high"
      IF new_priority IS NOT EMPTY THEN
        SET task.priority = new_priority
      ENDIF
    ENDIF
  END FOR

```

```

ENDIF

DISPLAY "Current Due Date: " task.due_date
REPEAT
    DISPLAY "Enter new due date (YYYY-MM-DD) (Press Enter to Skip): "
    GET new_due_date
    IF new_due_date IS EMPTY THEN
        BREAK
    ENDIF
    UNTIL validate_date(new_due_date) = TRUE
    IF new_due_date IS NOT EMPTY THEN
        SET task.due_date = new_due_date
    ENDIF

    DISPLAY "Task updated successfully!"
    RETURN
ENDIF
END FOR

DISPLAY "Task not found."
END FUNCTION

FUNCTION delete_task()
    CALL view_tasks()
    DISPLAY "Task Manager: Delete Task"

    REPEAT
        DISPLAY "Enter the task name to delete: "
        GET task_name
    UNTIL task_name IS NOT EMPTY

    FOR task IN tasks DO
        IF task.name = task_name THEN
            REMOVE task FROM tasks
            DISPLAY "Task deleted successfully!"
        ENDIF
    END FOR

    DISPLAY "Task not found."
END FUNCTION

DOWHILE TRUE
    DISPLAY "Task Manager: Menu"
    DISPLAY "1. Add Task"
    DISPLAY "2. View Tasks"
    DISPLAY "3. Update Task"
    DISPLAY "4. Delete Task"
    DISPLAY "5. Exit"

```

```
DISPLAY "Enter your choice (1-5): "  
GET choice  
  
IF choice = "1" THEN  
    CALL add_task()  
ELSE IF choice = "2" THEN  
    CALL view_tasks()  
ELSE IF choice = "3" THEN  
    CALL update_task()  
ELSE IF choice = "4" THEN  
    CALL delete_task()  
ELSE IF choice = "5" THEN  
    EXIT  
ELSE  
    DISPLAY "Invalid choice. Please enter a number between 1-5."  
ENDIF  
ENDDO  
END
```

Test Cases

Test Case 01

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 01	a	Invalid choice! Try again.	Checks whether the application handles nonnumeric input gracefully.	Invalid choice! Try again.	Pass

```
hasaldharmagunwardana@Hesxo iit % /usr/
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: a
Invalid choice! Try again.
```

Figure 1

Test Case 02

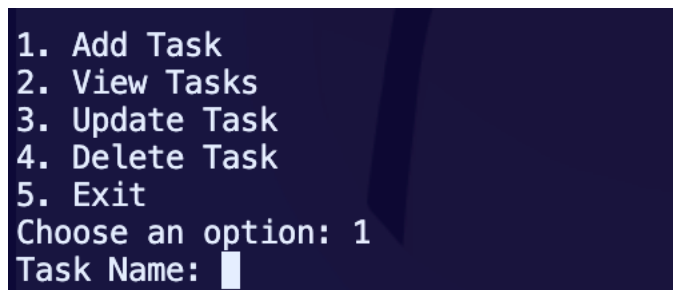
Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 02	9	Invalid choice! Try again.	Ensures the program rejects numeric input outside the valid option range.	Invalid choice! Try again.	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 9
Invalid choice! Try again.
```

Figure 2

Test Case 03

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 03	1	Prompt: Task Name:	Checks that selecting option 1 initiates the Add Task process.	Prompt: Task Name:	Pass



```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: █
```

Figure 3

Test Case 04

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 04	2	Prompt user again with error message	The input 2 is wrong because only "Low", "Medium", or "High" are allowed. The program showed an error and asked again.	Output: False Error message shown and user re prompted	Pass

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: Drink water
Description: Drink 5L water for Today
Priority (Low/Medium/High): 2
Invalid priority! Please enter Low, Medium, or High.
Priority (Low/Medium/High): 

```

Figure 4

Test Case 05

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 05	low	Program accepts valid priority after retrying and moves to due date prompt	User initially enters an invalid number for priority, then corrects it. Program proceeds.	Prompt: Due Date (YYYY-MM-DD):	Pass

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: Drink water
Description: Drink 5L water for Today
Priority (Low/Medium/High): 2
Invalid priority! Please enter Low, Medium, or High.
Priority (Low/Medium/High): low
Due Date (YYYY-MM-DD): 

```

Figure 5

Test Case 06

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
--------------	-------	-----------------	-------------	---------------	-------------

TC 06	a	False Show error and re prompt	Input a is invalid for due date. The system only accepts the format YYYY-MM-DD and must reprompted.	Output: False System shows format error and re-prompts	Pass
-------	---	--------------------------------	---	--	------

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: Drink water
Description: Drink 5L water for Today
Priority (Low/Medium/High): 2
Invalid priority! Please enter Low, Medium, or High.
Priority (Low/Medium/High): low
Due Date (YYYY-MM-DD): a
Invalid date format! Please use YYYY-MM-DD.
Due Date (YYYY-MM-DD): █

```

Figure 6

Test Case 07

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 07	2025-02-31	False Show error and re prompt	The format is correct, but February 31 is not a valid date. The system should reject it and re-prompt.	Output: False Error shown and re prompted	Pass

```

1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: Drink
Description: Drink 5L water for per Day
Priority (Low/Medium/High): low
Due Date (YYYY-MM-DD): 2025-02-31
Invalid date format! Please use YYYY-MM-DD.
Due Date (YYYY-MM-DD): █

```

Figure 7

Test Case 08

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 08	2025-07-28	Output: True Task added successfully	The third input 2025-07-28 is a valid date in YYYY-MM-DD format. July has 31 days, so 28 is valid. All other fields were valid too, so the task was added successfully.	Task Added Successfully!	Pass

```
Task Management System
Priority (Low/Medium/High): low
Due Date (YYYY-MM-DD): a
Invalid date format! Please use YYYY-MM-DD.
Due Date (YYYY-MM-DD): 2025-02-31
Invalid date format! Please use YYYY-MM-DD.
Due Date (YYYY-MM-DD): 2025-07-28
Task Added Successfully!
```

Figure 8

Test Case 09

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 09	2	True Show task details	User viewed tasks after adding one	Output: True Details displayed	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 2
1. Drink - Drink 5L Per day (Priority: Low, Due: 2025-02-27)
```

Figure 9

Test Case 10

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 10	3	True Show task details	User chose to update a task	Output: True Task list shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink - Drink 5L Per day (Priority: Low, Due: 2025-02-27)
Enter task number to update: |
```

Figure 10

Test Case 11

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 11	a	Output: False Show error message	User entered a letter	Output: False Invalid number shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink - Drink 5L Per day (Priority: Low, Due: 2025-02-27)
Enter task number to update: a
Invalid Task Number!
```

Figure 11

Test Case 12

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 12	2	Output: False Show error message	Only task 1 exists	Output: False Show error message	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink – Drink 5L Per day (Priority: Low, Due: 2025-02-27)

Enter task number to update: 2
Invalid Task Number!
```

Figure 12

Test Case 13

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 13	1	Output: True Prompt update fields	Valid task selected	Output: True Update started	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink – Drink 5L Per day (Priority: Low, Due: 2025-02-27)

Enter task number to update: 1
New Name (Enter to skip):
```

Figure 13

Test Case 14

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 14	Enter	Output: True Prompt next field	Skipped name update	Output: True Next prompt shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink – Drink 5L Per day (Priority: Low, Due: 2025-02-27)

Enter task number to update: 1
New Name (Enter to skip):
New Description (Enter to skip):
```

Figure 14

Test Case 15

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 15	Drink 10L Per Day	Output: True Prompt for priority	Description updated, next step shown	Output: True Prompt shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink – Drink 5L Per day (Priority: Low, Due: 2025-02-27)

Enter task number to update: 1
New Name (Enter to skip):
New Description (Enter to skip): Drink 10L Per Day
New Priority (Low/Medium/High) (Enter to skip):
```

Figure 15

Test Case 16

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 16	High	Output: True Prompt for due date	Updated priority	Output: True Next prompt shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 3
1. Drink - Drink 5L Per day (Priority: Low, Due: 2025-02-27)

Enter task number to update: 1
New Name (Enter to skip):
New Description (Enter to skip): Drink 10L Per Day
New Priority (Low/Medium/High) (Enter to skip): High
New Due Date (YYYY-MM-DD) (Enter to skip):
```

Figure 16

Test Case 17

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 17	2025-03-07	Output: True Task updated successfully	Updated all except name	Updated all except name	Pass

```
Enter task number to update: 1
New Name (Enter to skip):
New Description (Enter to skip): Drink 10L Per Day
New Priority (Low/Medium/High) (Enter to skip): High
New Due Date (YYYY-MM-DD) (Enter to skip): 2025-03-07
Task Updated Successfully!
```

Figure 17

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Test Case 18

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 18	2	Output: True Show updated task details	Viewed task after update	Output: True Details correct	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 2
1. Drink – Drink 10L Per Day (Priority: High, Due: 2025-03-07)
```

Figure 18

Test Case 19

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 19	4	Output: True Show task list to delete	Chose delete option	Output: True Task list displayed	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 4
1. Drink – Drink 10L Per Day (Priority: High, Due: 2025-03-07)
2. Run – Run 2KM For Day (Priority: Low, Due: 2025-03-03)
```

Figure 19

Test Case 20

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 20	3	Output: False Show error message	Chose non existing task	Output: False Error shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 4
1. Drink – Drink 10L Per Day (Priority: High, Due: 2025-03-07)
2. Run – Run 2KM For Day (Priority: Low, Due: 2025-03-03)

Enter task number to delete: 3
Invalid Task Number!
```

Figure 20

Test Case 21

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 21	1	Output: True Task deleted successful	Deleted existing task	Output: True Success message shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 4
1. Drink – Drink 10L Per Day (Priority: High, Due: 2025-03-07)
2. Run – Run 2KM For Day (Priority: Low, Due: 2025-03-03)

Enter task number to delete: 1
Task Deleted Successfully!
```

Figure 21

Test Case 22

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 22	2	Output: True Show remaining tasks	Viewed tasks after delete	Output: True One task shown	Pass

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 2
1. Run - Run 2KM For Day (Priority: Low, Due: 2025-03-03)
```

Figure 22

Test Case 23

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 23	1, Fasting, Don't Eat Food, High, 2025-03-07	Output: True Task added successfully	All inputs valid	Output: True Success message	Pass

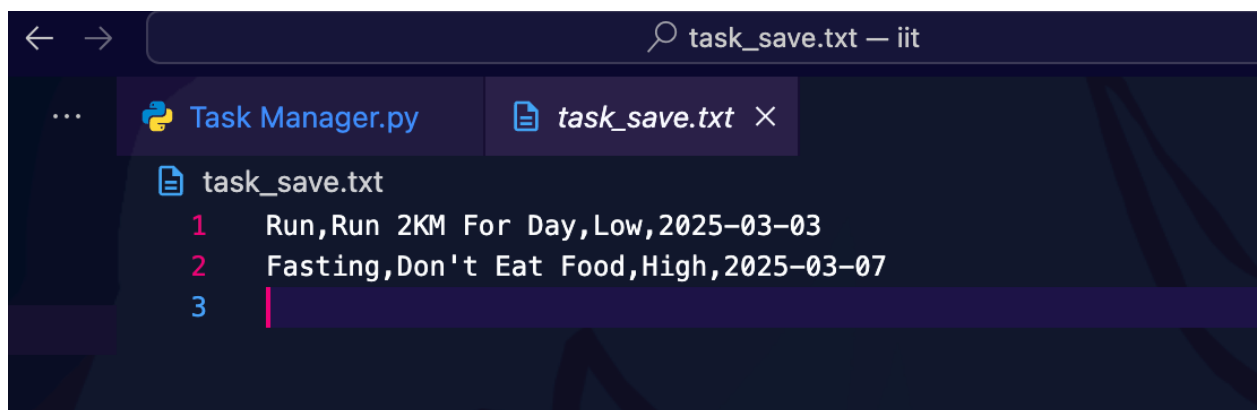
```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Exit
Choose an option: 1
Task Name: Fasting
Description: Don't Eat Food
Priority (Low/Medium/High): High
Due Date (YYYY-MM-DD): 2025-03-07
Task Added Successfully!
```

Figure 23

Test Cases for File Handling

Test Case 24

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 24	Tasks added via program	Output: True Tasks saved in task_save.txt	Verify file saving after adding	Output: True Data saved as CSV	Pass



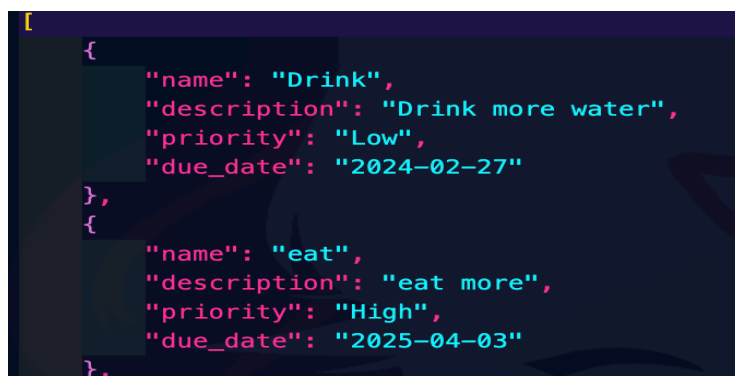
```
task_save.txt
1 Run,Run 2KM For Day,Low,2025-03-03
2 Fasting,Don't Eat Food,High,2025-03-07
3
```

Figure 24

Test Cases for JSON File Operations

Test Case 25

Test Case ID	Input	Expected Result	Description	Actual Result	Pass / Fail
TC 26	Tasks added via program	Output: True Tasks saved in tasks. Json	Save task and verify in file	Task saved correctly in JSON	Pass



```
[
  {
    "name": "Drink",
    "description": "Drink more water",
    "priority": "Low",
    "due_date": "2024-02-27"
  },
  {
    "name": "eat",
    "description": "eat more",
    "priority": "High",
    "due_date": "2025-04-03"
  }
]
```

Figure 25

Main Window

1. **Title Bar:** Displays the application name "Personal Task Manager – View Only" at the top of window.
2. **Search and Filter Panel:** this panel allows you to filter tasks by name, priority, and due date.
3. **Task Table:** Displays all your tasks in a tabular format with sortable columns.
4. **Sort Controls:** These buttons allow you to change the sorting order of tasks.



Search and Filter Panel

This panel contains the following components:

- **Task Name Field:** A text field where you can type keywords to filter the tasks by name.
- **Priority Dropdown:** A dropdown menu to filter tasks by specific priority level All, High, Medium, Low.
- **Due Date Field:** A text field where you can enter a specific date in YYYY-MM-DD format to filter tasks by due date.
- **Filter Button:** Applies the selected filters.

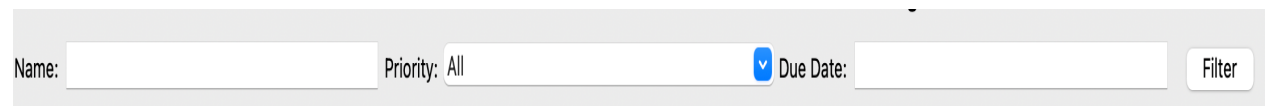
The image shows a search and filter panel with a light gray background. It contains four main input areas: a text field labeled 'Name:', a dropdown menu labeled 'Priority:' with 'All' selected, a text field labeled 'Due Date:', and a 'Filter' button on the right. The 'Priority' dropdown has a blue downward arrow icon.

Figure 27

Task Table

The table consist of four columns for each task:

- **Name:** The name of the task.
- **Description:** The detailed description of the task.
- **Priority:** The priority level (High, Medium, Low)
- **Due Date:** The deadline for the task in YYYY-MM-DD format.

Name	Description	Priority	Due_date
Study	file handling and functions	High	2025-04-18
run	run fast	Medium	2025-04-07
Gym	Chest day	Medium	2025-04-17
eat	eat more	High	2025-04-03
Drink	Drink more water	Low	2024-02-27
Clean	Remove old papers and organize files	Low	2025-04-21
Buy groceries	Milk, Bread	Medium	2025-04-20

Figure 28

Sort Buttons

The sort buttons allow you to organize tasks by:

- **Name:** Alphabetically sorts tasks by name ascending order.
- **Priority:** Sorts tasks by priority level (High → Medium → Low)
- **Due Date:** Sorts tasks by due date.

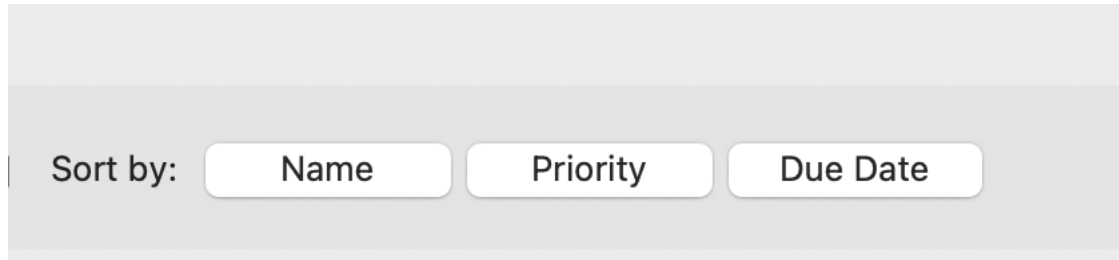


Figure 29

Click through the GUI

Launching the GUI

From the main menu, **select option “5” to Open the GUI interface**

```
hasaldharmagunwardana@Mac Task manger % /usr/local/bin/python3
"/Users/hasaldharmagunwardana/Desktop/iit /Task manger/Task Man
ager.py"
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Open GUI
6. Exit
Choose an option: 5
```

Figure 30

Filtering Tasks

1. By Name:

- Enter a keyword in the "Task Name" field.
- Click the "Filter" button.
- The table will display only tasks containing that keyword.

2. By Priority:

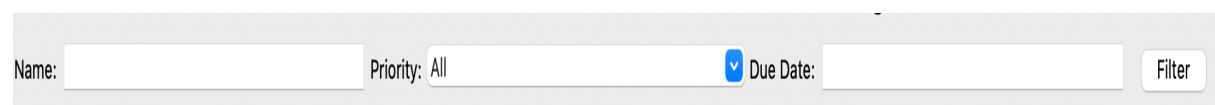
- Select a priority level from the dropdown menu.
- Click the "Filter" button.
- When "All" is selected, tasks of all priority levels are shown for other priority level it shows specific task with that priority.

3. By Due Date:

- Enter a date in YYYY-MM-DD format.
- Click the "Filter" button.
- Only tasks with that exact due date will be shown.

4. Combined Filtering:

- You can use any combination of the above filters.
- For example, enter a name keyword, select any priority, and enter a date.
- Click the "Filter" button.
- Only tasks matching filters will be shown.



The screenshot shows a horizontal filter bar with a light gray background. It contains three input fields: 'Name:' followed by a text box, 'Priority:' followed by a dropdown menu showing 'All', and 'Due Date:' followed by a date input box. To the right of these fields is a button labeled 'Filter'.

Figure 31

Sorting Tasks

1. Click any column header to sort by that column
2. Or use the sort buttons at the bottom:
 - Click "Name" to sort alphabetically by task name
 - Click "Priority" to sort by priority level (High → Medium → Low)
 - Click "Due Date" to sort chronologically

Name	Description	Priority	Due_date
Study	file handling and functions	High	2025-04-18
run	run fast	Medium	2025-04-07
Gym	Chest day	Medium	2025-04-17
eat	eat more	High	2025-04-03
Drink	Drink more water	Low	2024-02-27
Clean	Remove old papers and organize files	Low	2025-04-21
Buy groceries	Milk, Bread	Medium	2025-04-20

Figure 32

Exiting the Program

To close the Personal Task Manager, **select option “6”** from the menu. You’ll see a friendly **“Goodbye 🙋”** message, and the program will safely exit.

```
1. Add Task
2. View Tasks
3. Update Task
4. Delete Task
5. Open GUI
6. Exit
Choose an option: 6
Goodbye 🙋
hasaldharmagunwardana@Mac Task manger %
```

Figure 33